

Fixed Exchange Rates and Foreign Exchange Intervention

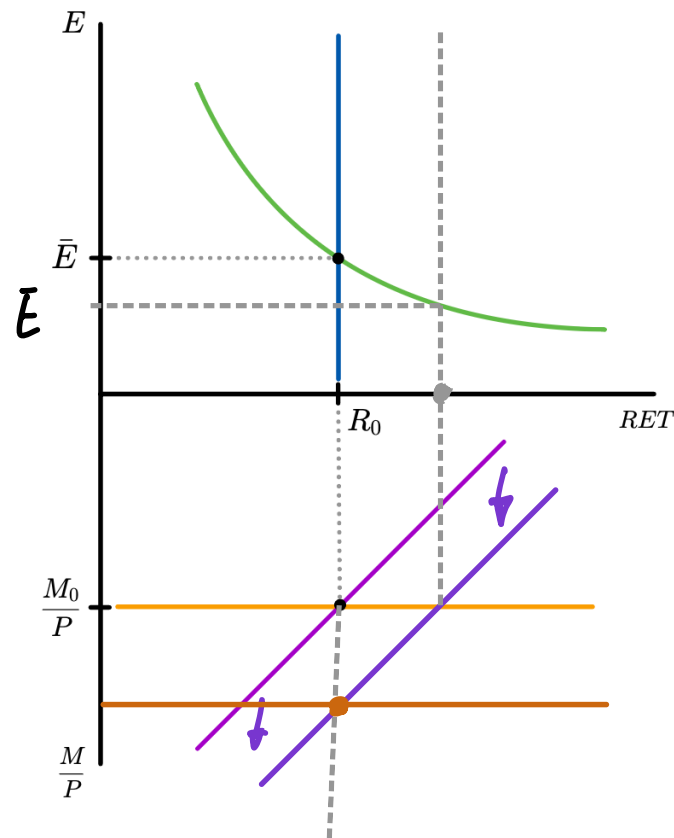
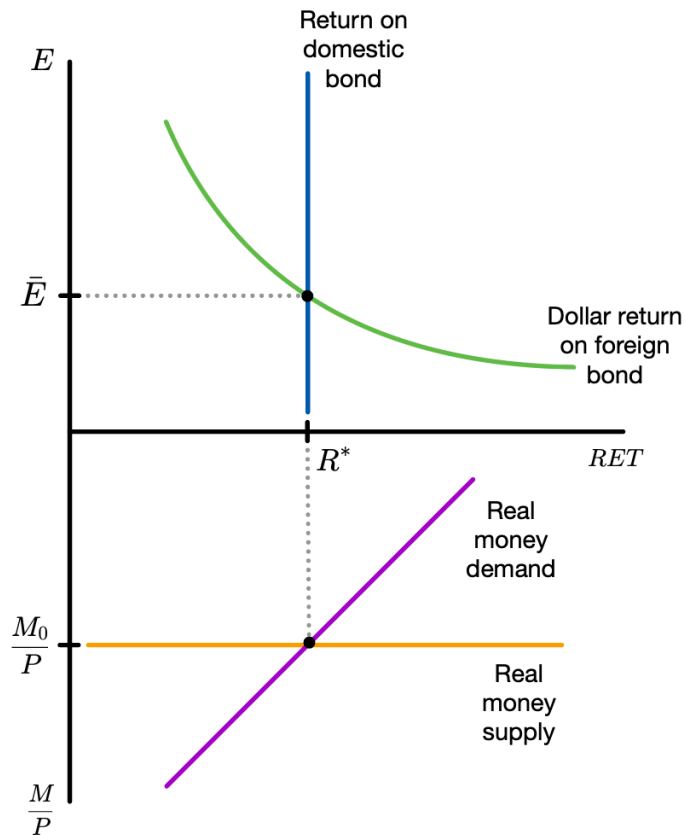
How the Central Bank Fixes the Exchange Rate

- Want to have a fixed exchange rate $E = \bar{E}$
- Assume $E = \bar{E}$ forever – the fixed exchange rate regime is said to be **credible**
- Then $E^e = E = \bar{E}$
- Central bank changes money supply so that $E = \bar{E}$ at all times
- UIP with $E^e = E$ implies $R = R^*$
- In a fixed exchange rate regime, a depreciation is called a **devaluation** and an appreciation a **revaluation**

Response to Changes in Y

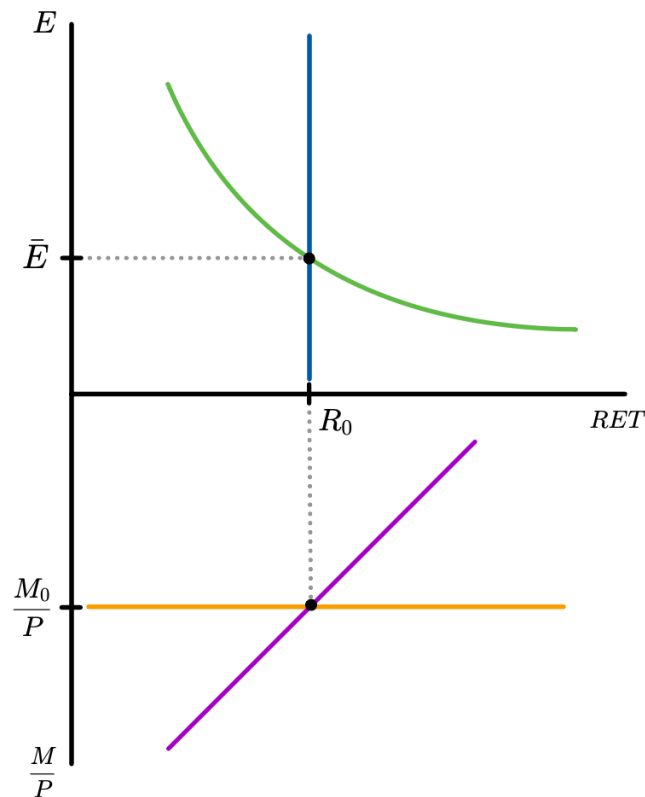
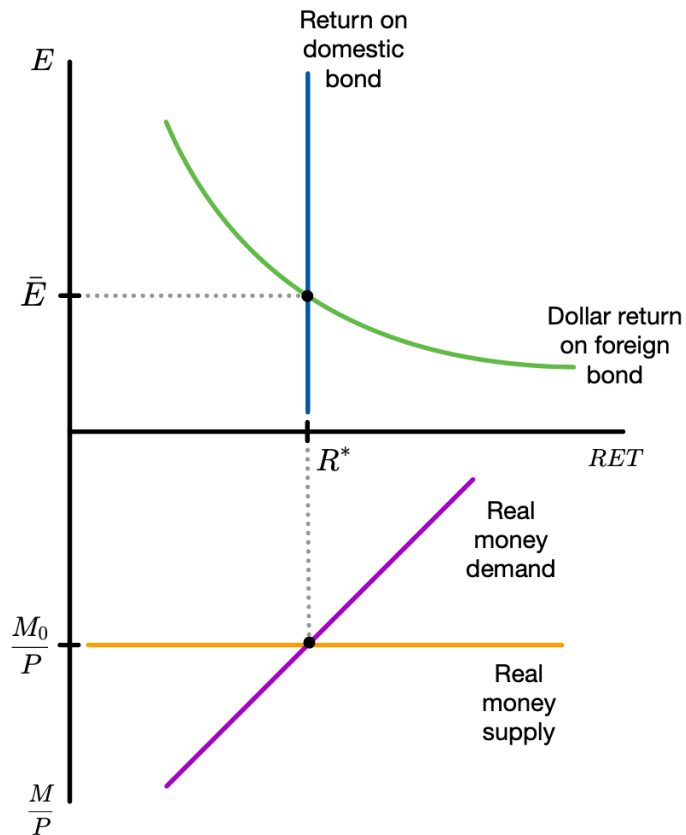
$Y \uparrow$

Y^* UNCHANGED



Response to Changes in R^*

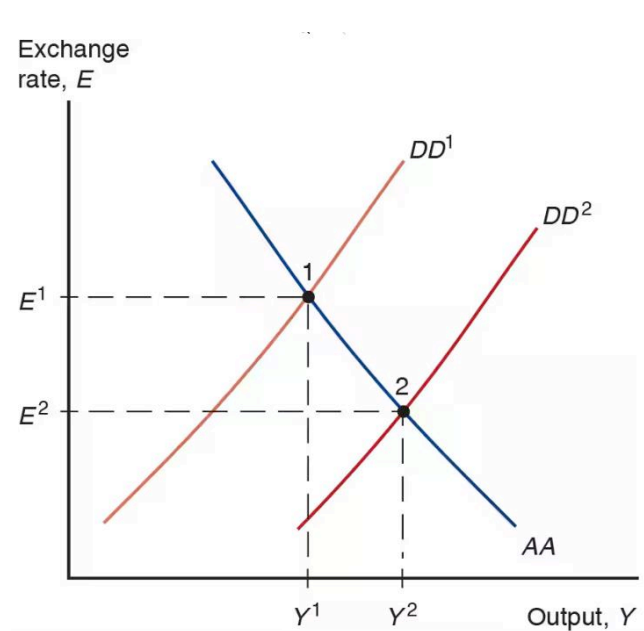
LEFT AS
EXERCISE :)



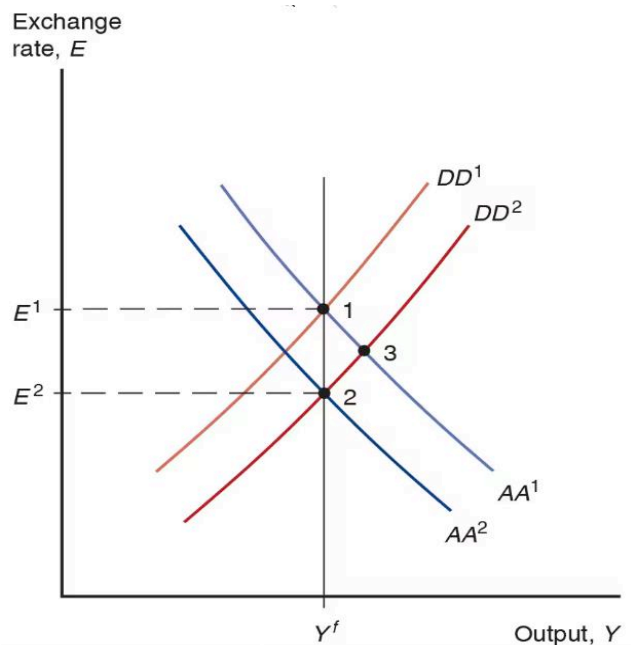
Fiscal Stimulus

Floating exchange rate

Temporary

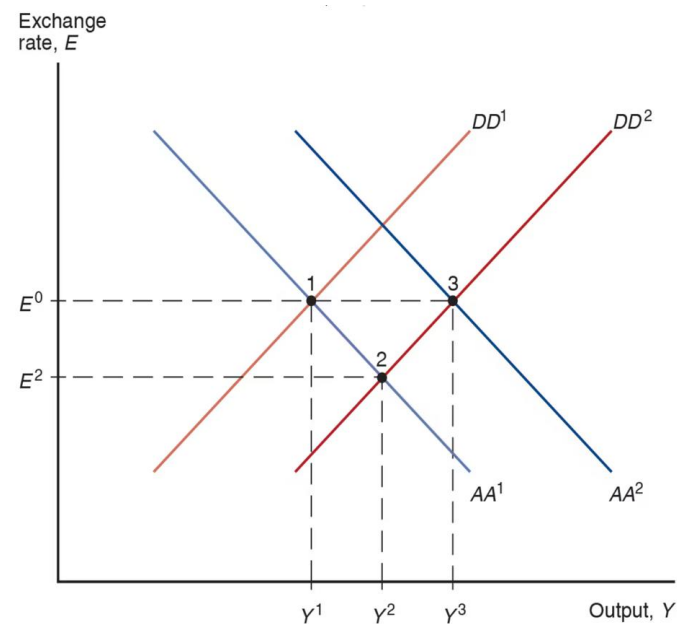


Permanent



Fixed exchange rate

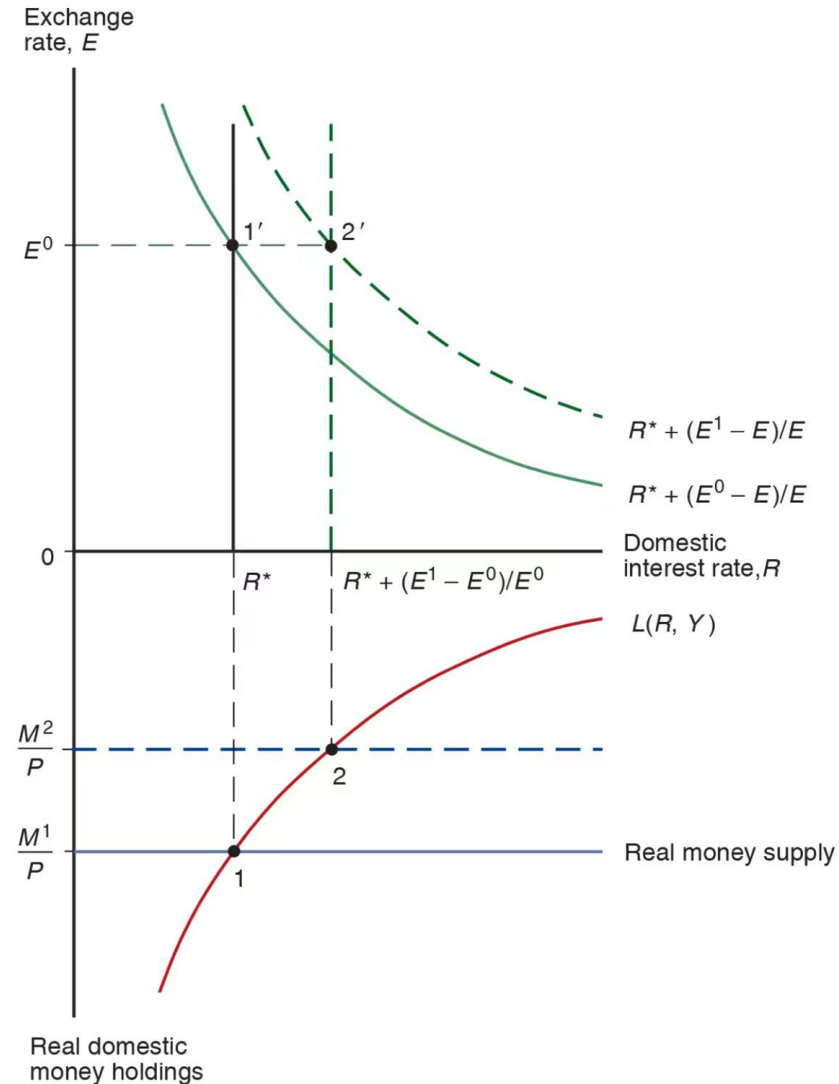
Temporary or permanent



Balance of Payments Crises

- Want to have a fixed exchange rate $E = \bar{E}$
- If $E^e = E = \bar{E}$ forever, the fixed exchange rate regime is **credible**
- Imagine E^e suddenly increases
- Without any government intervention, $E^e < \bar{E}$

- To maintain $E = \bar{E}$, money supply must decline from M^1 to M^2
- The domestic interest rate must increase to $R^* + \frac{E^1}{E^0} - 1$
- As money supply declines, R is always below the dollar return on the foreign bond
- Investors flee the domestic currency and domestic bond – a **capital flight**



Devaluation

- If central bank runs out reserves – or expects it will run out in the future – there will be a devaluation
- Fixed exchange rate must be modified or abandoned

